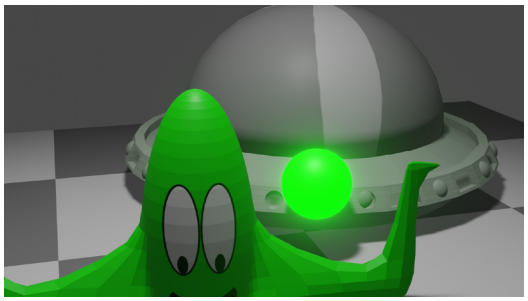
Lab: Advanced Animating

In the last lab you learned the basics of animating in Blender using the Blender Timeline and adding keyframes. In this lab we will expand on what we learned and animate a space scene.

Setup



Open *Lab-Space.blend* into Blender. You will see many 3D models in the scene including a spaceship and alien.

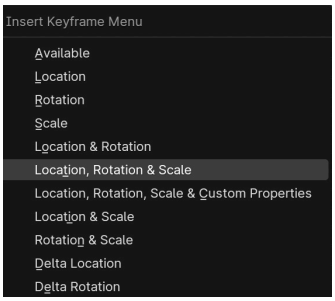
Make sure the Blender Timeline is at the bottom of your layout. If not, add a panel to the bottom of your Blender layout and in the *Editor Type* choose *Timeline*.

You can view the final animation below:

https://youtu.be/ULZToqdpdOg

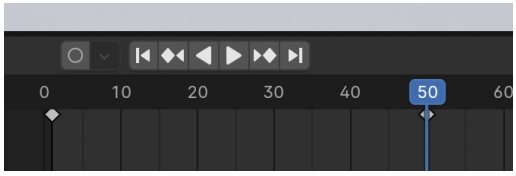
Animate the Alien

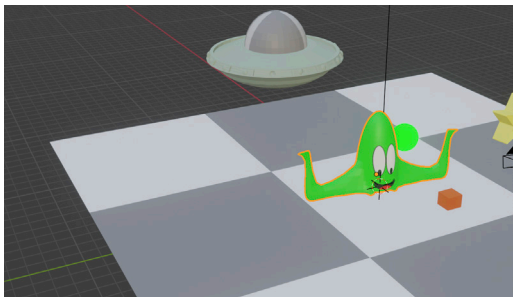
Let's do a little review from the last animation lab. If you view the reference animation the *Alien* moves across the scene, rotates and scales.



Move the blue time marker to the beginning of the *Timeline* or *frame 1*. Highlight the *Alien* model in the *3D Viewport* or the *Outliner*.

Use the *Move Tool* to move the *Alien* to location *(0,-15,1*), if is not already. Use the *Rotate Tool* to rotate the *Alien* to *(0,0,-66˚)*. Use the *Scale tool* to scale the *Alien* to (0,0, 3.15). Now that we set the location, rotation and scale let's set a keyframe for it. Press **"K" ("i" for Blender versions before 4.0)** on the keyboard to set a keyframe and choose the *Location, Rotation, Scale* option. This should create a diamond in the *Timeline*. Remember this is a keyframe or a saved point of data in the project.



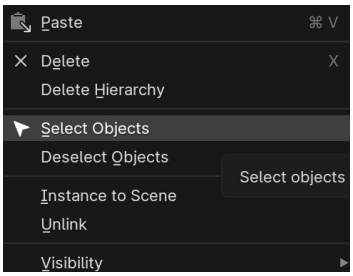


Move the time marker to *frame 50*. Use the *Move Tool* to move the *Alien* to *(0,0,0)*. Use the *Rotation Tool* to rotate the *Alien* to (0,0,180). Use the *Scale tool* to scale the *Alien* to (1.5, 1.5, 4.5) Press **"K"** again to set a second keyframe. A second diamond should appear in the timeline. Use the *Play* button in the timeline to play back the scene and you should see that the *Alien* moves and rotates across the scene.

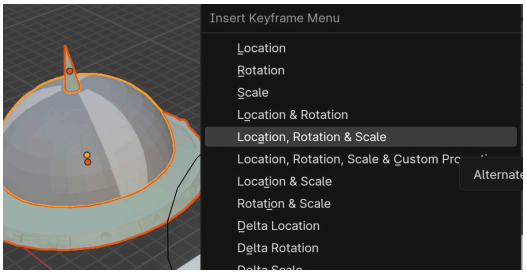
Complex Character

Animating a single mesh model like a cube or alien seems straight forward, but what if it is a complex model that is made up of many parts? If you look at the *Outliner* there is collection called *Spaceship*. This collection has several 3D meshes that are part of a

larger model of a spaceship. How do we animate this? There are two methods that we will demostrate. The first is you can select all of the items and animate each with the same keyframes.



Move the blue time marker to the beginning of the *Timeline* or *frame 1.* In the Outliner select the *Spaceship* collection. Right click and choose *Select Objects* from the pop up menu, this will be sure to select all of the objects in the collection. Press **'k'** to pull open the pop up menu and choose *"Location"*.



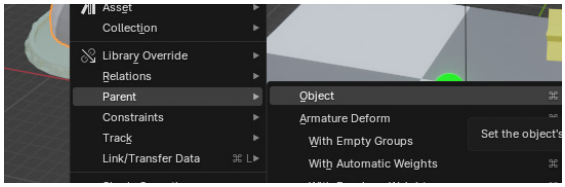
Move to *frame 50*. Make sure all of the objects are selected by using the *Select Objects* option. Use the *Move Tool* to move the group to the middle of the scene. Press **'k'** to set a new location keyframe. Save and scrub through the timeline, the spaceship should move across the scene.

Highlight and delete all of the keyframes you have made.

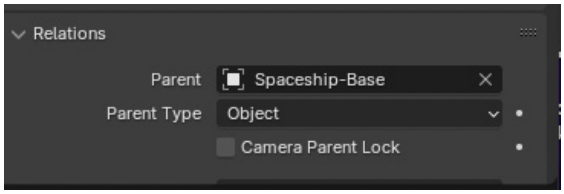
Parenting

Another way to handle this is parent the parts of the model to a main base element, and then animate that one base element. The benefit of this is setting less keyframes, which will make editing the animation easier in the future.

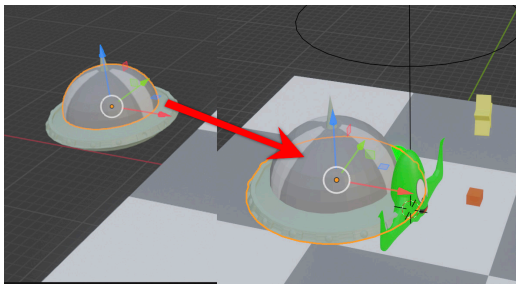
For this model let's make the *Spaceship* the main model. Let's begin by parenting the *Antenna* to the *Spaceship-Glass*. Select the *Antenna* model in the scene and then select the Spaceship-Glass, with parenting the order of selection is important so make sure that the Spaceship-Glass is the last selected model..



Go to *Object > Parenting > Object* to parent the *Antenna* to the *Spaceship-Glass*. The result of this is the *Antenna* mesh will be inside of the *Spaceship-Glass* mesh in the *Outliner*. Select the *Spaceship-Glass* and then select the *Spaceship-Base* and parent those objects. If you are unsure if an element is parented to another object you can go to *Object Properties > Relations > Parent.*



Move the blue time marker to the beginning of the *Timeline* or *frame 1.* Select the *Spaceship-Base* and press **'k'** to pull open the pop up menu and choose *"Location"*.

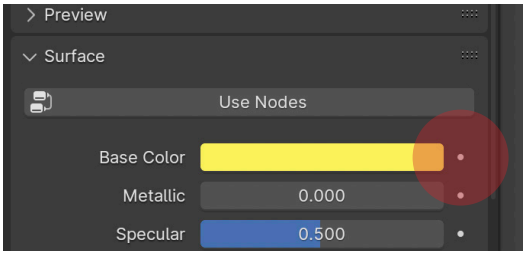


Move to *frame 50*. Use the *Move Tool* to move the group to the middle of the scene, position point *(-6,0,1.5)* and press **'k'** to set a new location keyframe. Save and scrub through the timeline, the spaceship should move across the scene.

Animating a Material

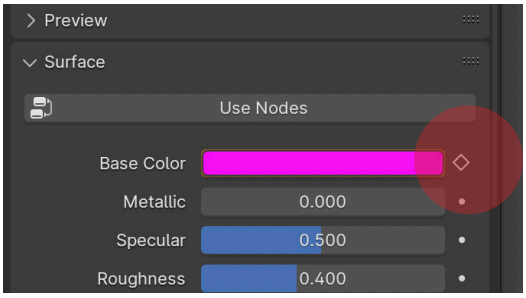
So, we know how to animate the location, rotation and scale for a single object, or a complex object; however there are many

things we can animate, including the materials on the model itself.

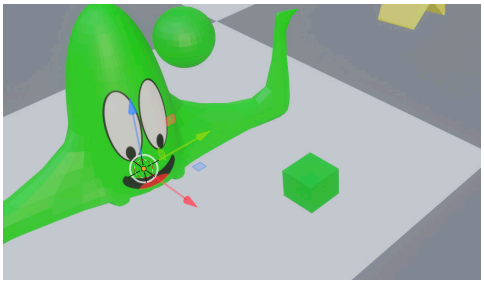


If you view the preview animation you will see that there is a star where the color animates from yellow to purple, let's try to recreate that.

Move the blue time marker to the beginning of the Timeline or *frame 1*. Select the *Star* model in the *3D Viewport* or *Outliner*. Click on the *Material Properties* and find *Base Color* under the *Surface* menu. You can see that the yellow has been set for the *Base Color* property. We can actually animate this property by clicking on the gray dot to the right of *Base Color* property. Clicking this will create a diamond keyframe in the timeline.



Move to *frame 50*. Change the *Base Color* from yellow to purple. Click on the diamond icon to the right of the *Base Color* and it will create a second keyframe.

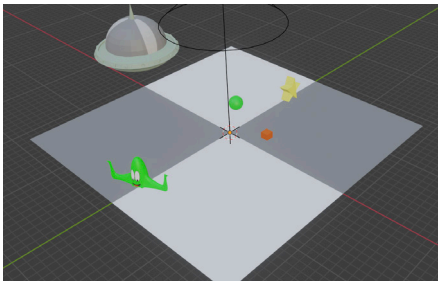


Scrub through the animation and you will see the star change from yellow to purple.

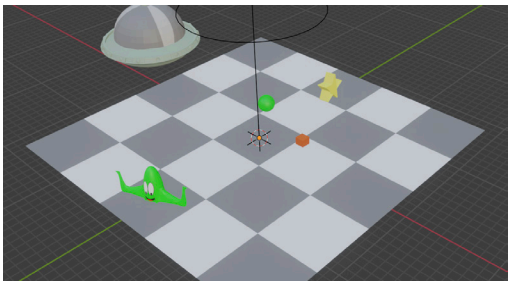
Repeat the process for the *Cosmic Cube*. Have the *Base Color* animate from red to green starting on *frame 1* and ending on *frame 50*.

Animated Texture

There are many properties in Blender that we can animate. If you view the preview you can see that the checker floor animates in terms of how many checkers there are. It goes from *2x2* to *10x10*, this can be done through keyframe animation.



Move the blue time marker to the beginning of the *Timeline* or *frame 1*. Select the *Ground* model in the *3D Viewport* or *Outliner*. If you click on the *Material Properties* you will see there is an option called *Scale*. This is how many many row / columns of checkers there are, currently it is *2* or 2x2. Click on the gray dot to the right of *Scale* to set a keyframe.



Move to *frame 50*. Change the *Scale* from *2* to *10*. Click on the diamond icon to the right of the *Scale* and it will create a second keyframe.

Scrub through the timeline and you will see that the number of rows / columns for the checker pattern animate.

Glowing Orb

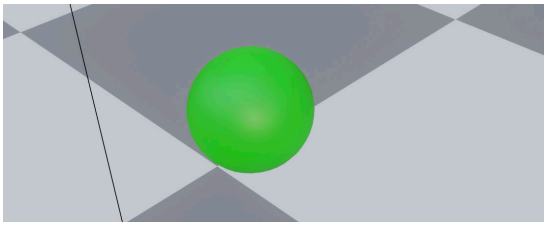
View the preview animation, you notice that there is a green orb that pulsates. We can use material properties to create this. Move the blue time marker to the beginning of the *Timeline* or *frame 1*. Select the *Orb* model in the *Outliner* and turn on the *Visibility*

toggle.

What sets the amount of *"glow"* from the orb is the *Strength* property in the *Emisson* subsection. On *frame 1* set the *Strength* to *7.0*. Click on the gray dot to the right of the *Strength* property to set a keyframe.

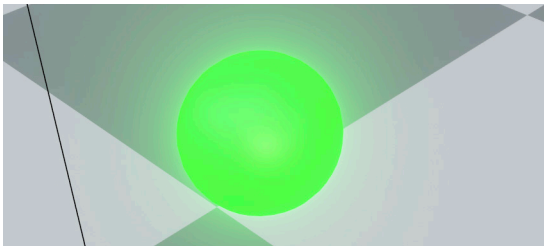


Move to *frame 20* and set the *Strength* to *0*. Click the diamond icon to set a keyframe.



Move to *frame 40* and set *Strength* to *7.0*. Set a keyframe.

Move to *frame 60* and set *Strength* to *0*. Set a keyframe.



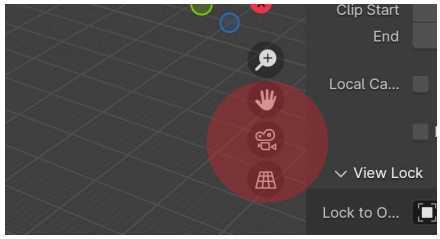
Move to *frame 80* and set *Strength* to *7.0*. Set a keyframe

Move to *frame 100* and set *Strength* to *0*. Set a keyframe.

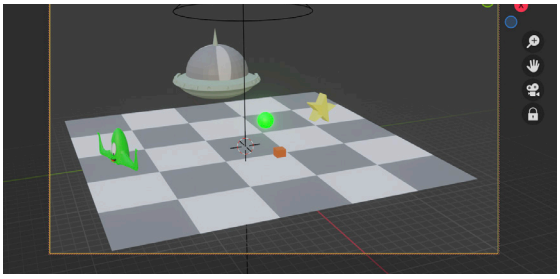
Scrub through the animation and you will see that the orb now has a glow that pulsates.

Animated Camera

In addition to animating a model or it's material properties we can also animate the camera for the project itself. This is useful for creating interesting shots like zoom or pans in our final animation.

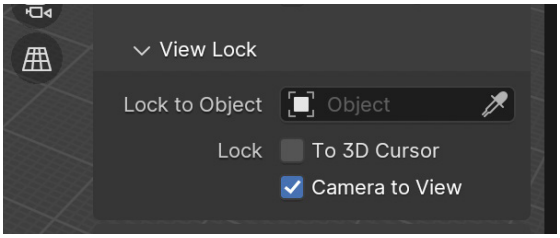


View the preview animation and you can see that the camera for the scene zoom closer to the *Spaceship* and the *Alien*. Let's try to recreate this.



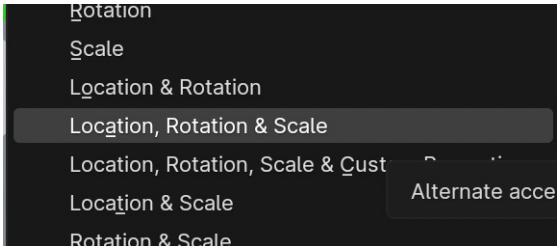
Move the blue time marker to the beginning of the Timeline or *frame 1* and select the *Camera* from the *Outliner*. To view the scene from the camera's view click on the *Toggle the Camera View* icon at the top right of the *3D Viewport*. Your scene should look something like the image on the right.

Let's set the camera to follow the navigation commands like orbit and pan. Press **"n"** to open the side panel. Under *View* click on *Lock > Camera to View*. This will allow you to edit the camera's location using the zoom and pan tools.



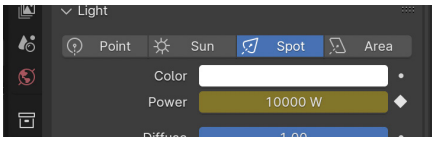
Set the camera to where you want the animation to start. You should be zoomed out fairly far. When you have set your camea view get out of the camera view by clicking on *Toggle the Camera View* again. With the camera object selected in the scene press **"k"** and set a *Location, Rotation, Scale* keyframe. This will set a keyframe.

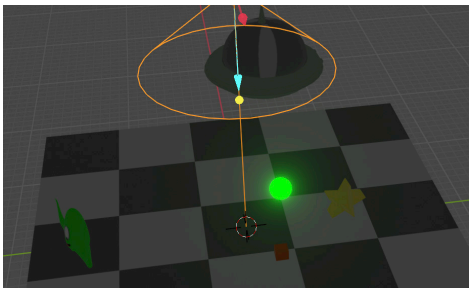
Move to frame 100. Return to camera view by clicking on the *Toggle the Camera View*. Zoom into the scene so the camera is zoom in on the *Spaceship* and *Alien*. Exit camera view and press **"k"** to choose Location, Rotation, Scale keyframe.

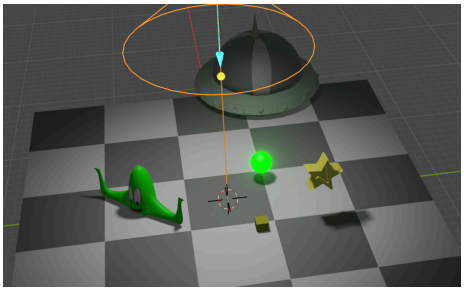


Return to camera view and scrub through the animation and you should see that the camera has now been animated.

Animated Light







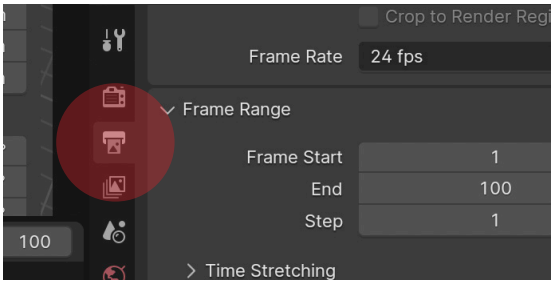
In addition to animating cameras in a Blender scene you can also animate lights as well. This allows to animate features such as the brightness or intensity, location of the light and even the color of the light.

If you view the animation preview you will see that the scene darks dark, and then fades in with a brighter light, let's try to create this.

Move the blue time marker to the beginning of the Timeline or *frame 1* and select the *Spot Light* from the *Outliner*. Select the *Light* property icon and find the *Power* option. Set the *Power* option to *0w* or off and press the gray dot to the right of the Power to set a keyframe.

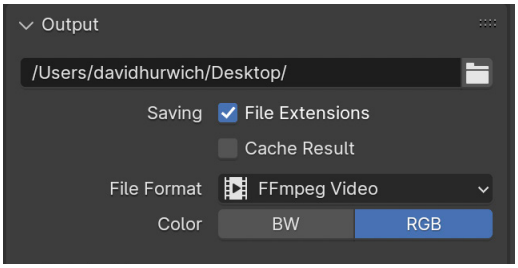
Move to *frame 25*. Change the *Power* from *0w* to *10,000 W* and set a second keyframe using the diamond. Play the scene back in *Rendered* mode and you will see that the light in the scene fades in.

Rendering



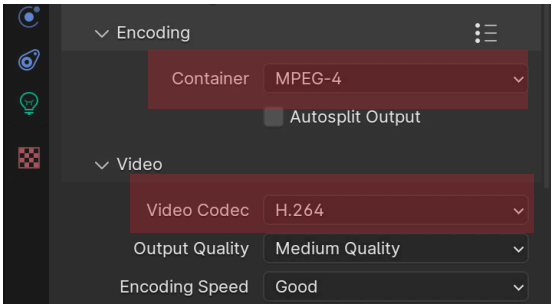
Now that we've animated the scene, let's export our animation into a video format. This will allow us to upload our animation to a video sharing service like YouTube or Vimeo or import into a video editing program like Adobe Premiere.

Before we render our scene, let's double check we have the settings we want. Click on the *Output* tab in the *Properties Panel*. Double check the frame rate and resolution of the animation. You should export the animation at *24fps* and at HD format which would be *1920 x 1080px.*



Next set where the animation will be output to, this can be done in the *Output* section. I recommend setting it to the *Desktop* of your computer.

Finally set the color mode and *File Format*. Set the *Color Mode* of the video to *RGB* and set the *File Format* to *FFmpeg* Video. Under *Encoding* change it from *Matroska* to *MPEG-4*. Under View make sure the *Video Codec H.264* is being used.



Go to *Render > Render Animation* and render out your final animation.

Submission

After exporting your space animation upload it to a video sharing service such as YouTube or Vimeo. After uploading copy the url to the video and paste it into a text document. Save the document as *Space-Lab-URL*.

Submit your *Lab-Space.blend* file and *Space-Lab-URL* document in a zipped compressed folder to Canvas for grading.